Can a compressed alert still surprise us?

Fingerprinting LSST alerts, and a test with circles and squares

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RTF: Real-Time Fingerprints

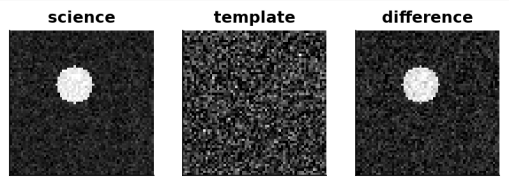
October 2026

The problem: ten million alerts a night

- LSST will issue ∼10 million alerts every night.
- Each one: the light curve so far, a 63×63 pixel cutout triplet, and catalogue metadata.
- Brokers filter the stream down to what they can serve and discard the rest.
- What you did not know to look for never reaches you.

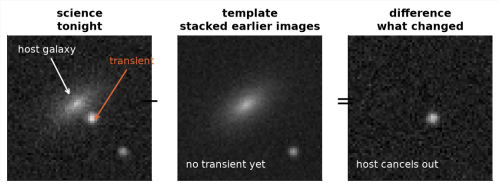
raw stream ∼100 MB/s

what can be served ∼1 MB/s (one HD video)



synthetic cutout triplet used later in this talk

What is in an alert, 1/3: the cutout triplet

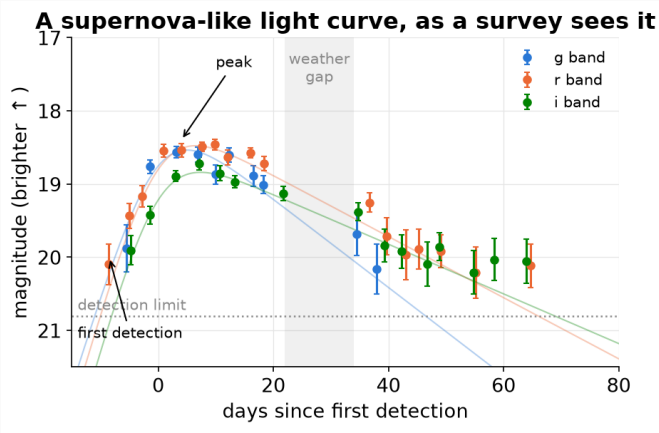


- **Science:** tonight's exposure, 63×63 pixels ($\approx 1'$) around the candidate: transient, host, neighbours, sky.
- **Template:** a deep stack of earlier images of the same field. The sky *before*.
- **Difference:** the survey pipeline's subtraction. What changed.

Why all three?

- The difference is *not* science minus template of these crops: the pipeline makes it on full frames with PSF matching and resampling the crops cannot reproduce.
- The template gives host context: nuclear, offset, on a star, empty sky.
- Science + template together expose bad subtractions.

What is in an alert, 2/3: the light curve



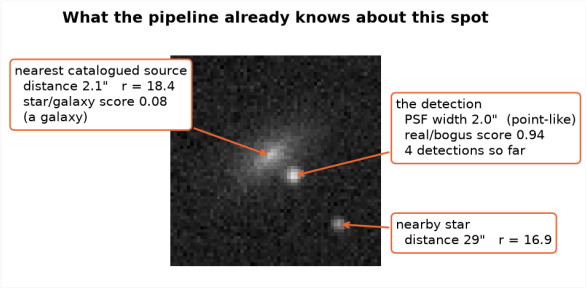
Brightness vs. time in three filters, sampled whenever the survey happened to look. Rise time, decline rate and colour evolution separate the classes.

What the model receives, per detection:

time since first detection	$\log(1+\Delta t)$
gap since previous one	$\log(1+\delta t)$
flux, flux error	log
filter	g / r / i flag

A variable-length list of such rows; the two time features exist because the sampling is irregular. In today's toy this input is switched off.

What is in an alert, 3/3: the metadata

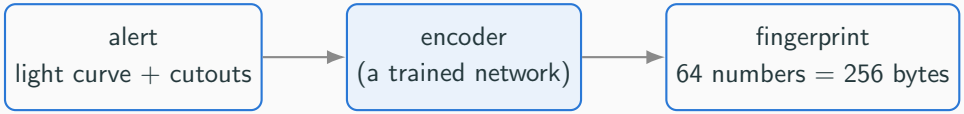


About 30 numbers the pipeline has already computed for every alert, mostly about *context*:

- nearest catalogued source: distance, brightness, star or galaxy
- how point-like the detection is
- the survey's own real-vs-bogus score
- how many detections so far

The pixels say what it looks like; the metadata says what the catalogues already know about that spot. Also switched off in today's toy.

The idea: a 256-byte fingerprint per alert



The analogy you already use

- A full spectrum $\rightarrow$ (colour, magnitude) on a CMD.
- Two numbers, chosen by physics, that preserve what matters.
- A fingerprint is the same idea with more numbers, chosen by training on 18 000 labelled ZTF transients.

What you can do with fingerprints

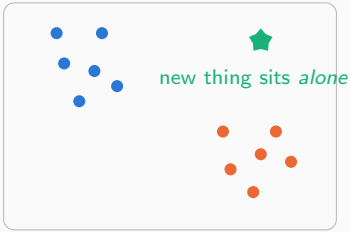
- Classify: 87 % accuracy from 256 bytes (full pipeline: 94 %).
- Find look-alikes: nearest neighbours in fingerprint space.
- Flag outliers: things far from everything seen before.

The tension: recognise vs. discover

trained hard to recognise

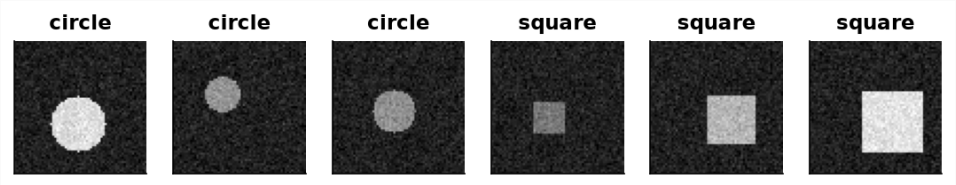


keeps more than it needs



- The network keeps only what its training rewards.
- If the training question is “SN Ia or TDE?”, nothing rewards keeping “this looks like neither”.
- This is the question the rest of the talk tests.

Why a toy: circles and squares



- Before trusting this on real transients, test the image part alone on a problem whose answer we know.
- Light-curve input switched off: the fingerprint depends only on the cutout.
- Synthetic stamps: circles and squares of equal area, random size, position and brightness, plus noise. *Nothing but shape distinguishes them.*
- If the fingerprint cannot separate these, nothing downstream is meaningful.

Two ways to build the image encoder

From scratch

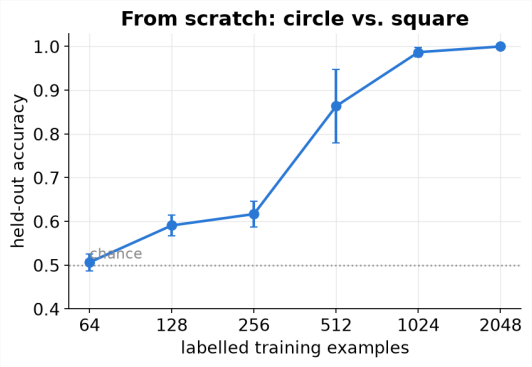
0.25 million parameters
starts knowing nothing
learns only from our stamps

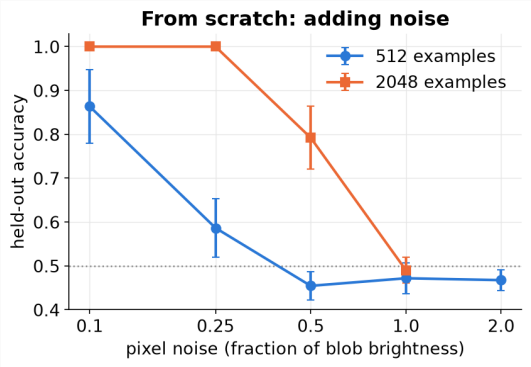
Zoobot

8.5 million parameters
pre-trained on Galaxy Zoo morphology votes
weights frozen: used exactly as published

- Zoobot is a public model (Walmsley et al.). We never retrain its weights; we only train a small read-out that maps its output into our 16 numbers.
- A first-year student vs. someone who has classified galaxies for a decade. Both get the same homework.

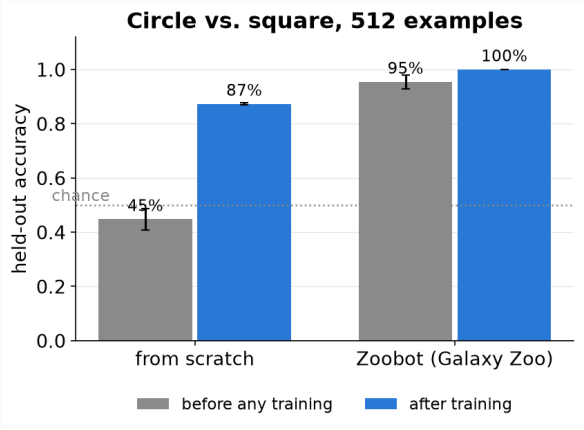
Result 1: from scratch works, but is hungry and fragile





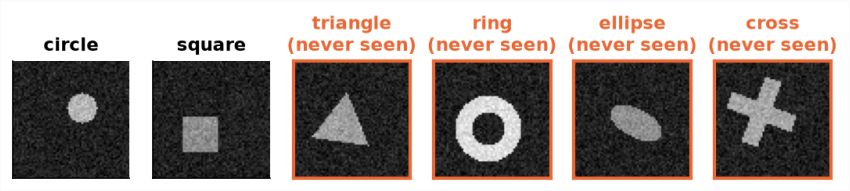
- 256 examples: 62 %. About 1000 examples: 99 %.
- Add modest noise and 512 examples is no longer enough; 2048 fixes it.
- Rare transient classes do not come with 1000 labelled examples.

Result 2: Zoobot already knows



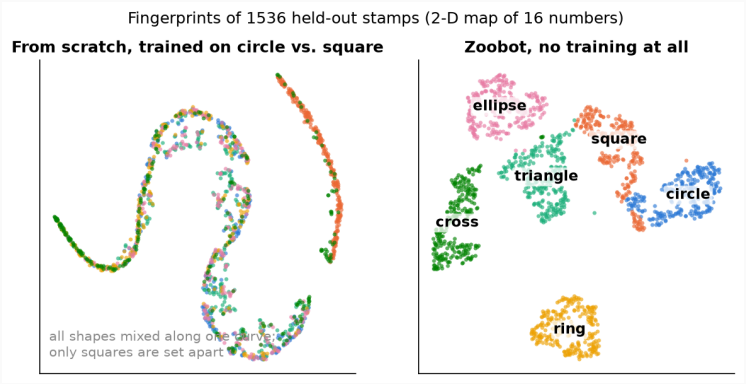
- "Before" means a random read-out on top of frozen Zoobot features. No training on circles or squares whatsoever.
- It already separates them at 95 %.
- This is the data-efficiency argument in one picture.

The discovery test: shapes it has never seen



- Train on circles and squares only. Then show four new shapes and ask, *without telling it*:
  1. Is this fingerprint far from every circle and square we trained on? (score: how often a new shape looks more unusual than a known one; 50 % useless, 100 % perfect)
  2. If we group the fingerprints blindly, do the new shapes get their own groups?
- Ceiling check: *if* we had labels, could a straight line separate all six?

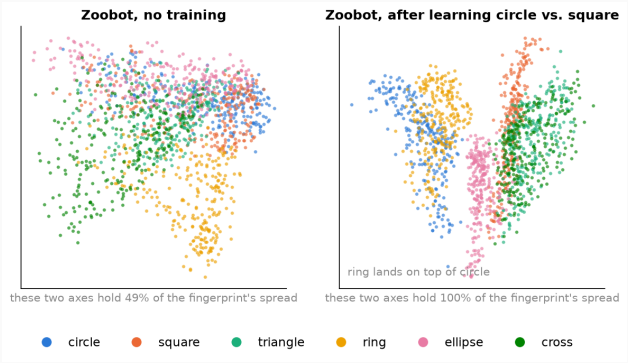
Result 3: from scratch cannot; Zoobot can



From scratch: new-shape detection 68 %; six-shape separability 43 %. Triangles, rings and ellipses are all filed as "circle", confidently.

Zoobot, no training: detection 99.8 %; separability 99.5 %. Ring and cross form their own islands; triangle and ellipse sit close.

The catch: teaching it to classify squashes the fingerprint



- Once the read-out learns circle vs. square, the fingerprint collapses onto a line.
- Six-shape separability: 99.5 % → 85 %. Ring now sits on top of circle.
- New-shape detection holds up (96 %): each shape keeps its own stretch of the line.
- Classification rewards one direction and lets the others fade. Adding "reconstruct the image" recovers most of it (97 %).

What this means, and what is next

Three takeaways

1. Use a Galaxy Zoo pre-trained encoder for cutouts. From scratch is not viable for rare classes.
2. Be deliberate about how hard the fingerprint is pushed to classify; keep a reconstruction-type objective in the mix.
3. Measure discoverability on real alerts next: train without TDEs, then ask whether TDEs look unusual.

Honest caveats

- Synthetic shapes are easy. Real stamps have hosts, PSF blur and faint sources.
- Not yet tested: whether an ordinary ImageNet-pre-trained network does as well. If it does, the win is "pre-training", not "Galaxy Zoo".

Backup: the seven fingerprint variants

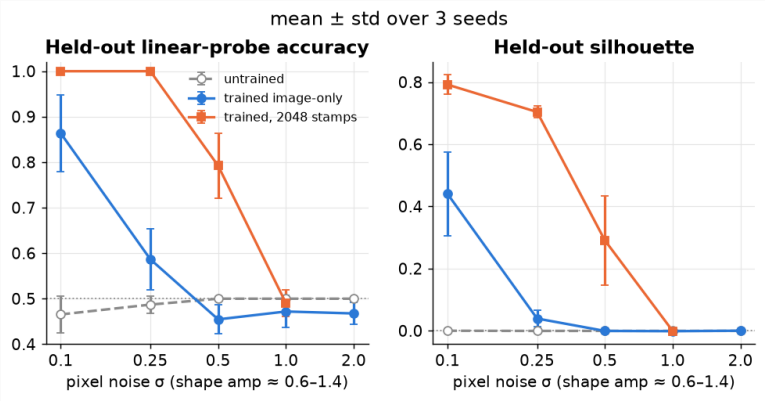
fingerprint built by	separability (labels)	new-shape detection	blind grouping	dims used
untrained small network	0.23	0.55	0.00	3.3
small network, classify	0.43	0.68	0.28	1.0
small network, reconstruct	0.34	0.75	0.01	5.2
small network, both	0.44	0.68	0.23	2.7
Zoobot, no training	**0.995**	**0.998**	0.54	4.8
Zoobot, classify	0.85	0.96	0.54	1.0
Zoobot, both	0.97	0.97	**0.62**	1.4

2048 training stamps, 256 held-out stamps per shape, 2 seeds (Zoobot untrained: 1).

Backup: how the numbers are defined

- **Separability (with labels):** 6-way logistic regression on the 16-number fingerprint, 5-fold cross-validated accuracy.
- **New-shape detection:** mean distance to the 5 nearest circle/square training fingerprints; AUROC of new vs. known.
- **Blind grouping:** k-means with k=6; adjusted mutual information against the true shape.
- **Dims used:** participation ratio of the fingerprint covariance (16 = all directions equal, 1 = a line).
- **The 2-D maps:** PCA shows the two largest directions (honest, can hide structure); t-SNE folds all 16 into the plane (shows structure, exaggerates gaps).

Backup: is noise a visibility problem or a data problem?



Blob brightness is 0.6–1.4. With 2048 examples the from-scratch network is perfect at noise 0.25 and still useful at 0.5; only at noise ≈ brightness does more data stop helping.